

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 8: Principal Component Analysis (PCA)

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1 Keywords

- Covariance matrices, dimensionality reduction, PCA, Duality

2 Main Points

2.1 PCA

- This is revision as you should have all seen this in foundations of ML
- The covariance matrix is defined as

$$\mathbf{C} = \frac{1}{m-1} \sum_{i=1}^m (x_i - \hat{\mu})(x_i - \hat{\mu})^\top$$

- Defining the matrix $\mathbf{X}$ as

$$\mathbf{X} = \frac{1}{\sqrt{m-1}} (x_1 - \mu, x_2 - \mu, \dots, x_m - \mu)$$

then $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$

- The *principal components* are the eigenvectors of the covariance matrix with the largest eigenvalues
- We can reduce the dimensionality of the inputs by projecting into the subspace spanned by the principal components
- We can reconstruct a vector from its principal component projection

$$\hat{x} = \sum_i z_i v_i$$

- v_i are the principal components (eigenvectors of the covariance matrix with largest eigenvalues)
 - $z_i = v_i^\top x$ are the values of the new features
 - the sum is over the principal components that we
- The expected squared error reconstruction loss $\mathbb{E}[(\hat{x} - x)^2]$ is equal to the sum of the eigenvalues we ignore

2.2 Duality

- We can define the dual matrix, $\mathbf{D} = \mathbf{X}^\top \mathbf{X}$ with components $D_{kl} = (\mathbf{x}_k - \boldsymbol{\mu})^\top (\mathbf{x}_l - \boldsymbol{\mu})$
- If $\mathbf{v}_i$ is an eigenvector of $\mathbf{D}$ with eigenvalue λ_i then $\mathbf{u}_i = \mathbf{X} \mathbf{v}_i$ is an eigenvector of $\mathbf{C}$ with the same eigenvalue
- Matrix $\mathbf{C}$ and $\mathbf{D}$ have exactly the same non-zero eigenvalues
- If we have more features than training examples it is more efficient to work with $\mathbf{D}$ than $\mathbf{C}$
- Note in this case the training examples will not span the feature space. $\mathbf{D}$ describes the fluctuations in the space spanned by the examples

3 Exercises

3.1 Duality

- Show that if $\mathbf{v}_i$ is an eigenvector of $\mathbf{D}$ with eigenvalue λ_i then $\mathbf{u}_i = \mathbf{X} \mathbf{v}_i$ is an eigenvector of $\mathbf{C}$ with the same eigenvalue
- Answer in the lecture notes

4 Experiments

4.1 Duality

- Using Matlab/Octave or python illustrate that the dual matrix and covariance matrix have the same eigenvalues

```
X = randn(5,3)    % construct a random matrix
C = X*X'           % compute a sort of covariance matrix (haven't bothered removing mean
D = X'*X           % compute dual
[V,LC] = eig(C)    % compute eigensystem of C
[U,LD] = eig(D)    % compute eigensystem of D should have the same non-zero eigenvalues
u1 = X*U(:,1)      % left multiply and eigenvector of D by X
u1/norm(u1)        % normalise above should be the same as V(:,3) (could be V(:,4) or V(:,5))
```